Management Equity Incentives and Tone Management of Earnings Communication Conferences: *Evidence from China*

1 Institutional background

1.1 Earnings communication conferences

Since 2005, earnings communication conferences have been crucial for investor relations management among Chinese listed companies. The China Securities Regulatory Commission (CSRC) issued the *Guidelines on Investor Relations Management for Listed Companies* in April 2022, requiring timely conferences post-annual report disclosure, per CSRC and Stock Exchange regulations. To ensure order and transparency, the Shanghai and Shenzhen Stock Exchanges have set guidelines. Listed firms must publish advance notices of conference details, invite key personnel (e.g., the chairman, finance head, strategic development representatives), and record and publish post-conference investor relations activities.

In recent years, the number of listed companies holding these conferences has grown significantly. In 2021, 3,756 companies, or 87.41% of those disclosing annual reports, held earnings communication conferences. This development not only provides a historical context for examining the effectiveness and influence of earnings communication conferences but also offers empirical data to support this research.

1.2 Management equity incentives

According to the regulation in China, namely the Administration of Equity Incentives in Listed Companies, it is clarified that "equity incentive is long-term incentives provided by a listed company to its directors, senior management, and other employees based on its own stock." The Company Law stipulates the objects of equity incentives, including directors, management, core technical personnel, core business personnel, and other employees who have a direct impact on the company's performance and future development. As the equity incentive plan will not only need to explain the purpose and object of the equity incentive, but also need to clarify the type of incentive, exercise conditions, and exercise period. Today, the mainstream equity incentive methods for listed companies in China are restricted stocks and stock options. Whether a listed company adopts the incentive type of granting restricted stocks to determine the grant price, or the incentive type of granting stock options to purchase stocks at a predetermined exercise price when rights of equity incentive can be exercised. The unlocking condition is that the interval between the grant date of the stock and the exercise date is greater than 12 months, and it needs to be unlocked in installments, with each period not less than 12 months. In addition, it is worth mentioning that after the end of each exercise period, the current stock options of the incentive object that have not

been exercised should be terminated, and the listed company should cancel them on time. If the grant price of equity incentives is low or the stock price is high when unlocked, the management will receive higher returns in the future. This shows that the management has the motivation to manipulate the stock price at both times.

2 Definitions for all variables

Table I Measurements of the variables (full version)

Type	Variables	Measurements
Main variables	Ifincent	An indicator variable set to 1 if the equity incentives grants, and 0 otherwise
	Incrate	Divide the number of incentive stocks granted by the total number of shares
	Lnvalue	Ln (the number of granted equity incentives × the close price of the granted date+1)
	Medab	Median of the abnormal tone of earnings communication conferences response, calculated by equation (2)
Control variables	Sep	The difference between the actual controller's control and ownership
	Roa	Net profit divided by the total assets
	Droa	The return on total assets for the current year minus the return on total assets for the previous year.
	Ret	Annual stock return calculated by $[(P_t - P_{t-1}) + Div_t]/P_{t-1}$, where P_t is the stock close price in year t, and P_{t-1} is the stock close price in year t-1
	Stdroa	The standard deviation of the return on total assets over the past five years.
	Loss	An indicator variable set to 1 when net profit is negative, and 0 otherwise
	Seo	An indicator variable set to 1 if the company conduct seasoned equity offering, and 0 otherwise
	Soe	An indicator variable set to 1 for state owned enterprise, and 0 otherwise
	Age	Ln (1+years since IPO)
	Size	Ln(total assets)
	Btm	The book value of equity divided by market value of equity
Robust check	Meanab	Mean value of the abnormal tone of earnings communication conferences response, calculated by equation (2)
	Incarea	The equity incentive coverage rate in the province where the company is located , calculated by the total number of enterprises implementing equity incentives in the province divided by the total number of enterprises in the province
Further discussion	Incenstype	Stock options granted are assigned a value of 1, whereas restricted stock is assigned a value of 0.
	Execut	An indicator variable is set to 1 if exercisable within the current year, and 0 otherwise.
	Exerat	The ratio of granted incentive stocks to the total number of shares.
	Exvalue	Ln (the number of exercisable options × the exercise price + 1)
	Ifcut	An indicator variable set to 1 if executives engaged in stock option exercise in the current year for equity incentives, and 0 otherwise
	Lncutm	Ln (the number of shares sold by executives exercising stock options for equity incentives + 1)
	Lncutv	Ln (the total market value of shares sold by executives exercising stock options for equity incentives + 1)
	Medtone	Median of the tone of earnings communication conferences response, calculated by equation (1)

Pospct	The number of positive words divided by the total number of words of the conference
Negpct	The number of negative words divided by the total number of words of the conference
Roe	Net profit divided by the stockholder's equity
Abseffic	Inefficient investment following Richardson (2006), calculated by the difference between actual investment expenditure and expected investment expenditure, and then takes the absolute value
Rm	The real earnings management following Cohen and Zarowin (2010).
Da	The accrual-based earnings management measured as the discretionary accruals following Kothari *et al.* (2005).
Gap	Derived from dividing the median salary of the top three executives in peer companies within the industry. Peer companies refer to those with sales revenue either exceeding or falling below the industry median.
Eecomp	The difference between actual executive compensation and expected executive compensation following Cai and Walkling (2011).
Newsneg	The number of negative reports divided by the number of media reports

Notes: This table displays the definition of variables.

3 The tables from Section 5.2

Table XII The influence mechanism of equity incentives on positive and optimistic tone

	Whether to grant equity incentives			The number of equity incentives granted			The market value of equity incentives granted		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	F.Medtone	*F.Roe*	*F.Medtone*	*F.Medtone*	*F.Roe*	*F.Medtone*	*F.Medtone*	*F.Roe*	*F.Medtone*
F.Roe			0.132***			0.133***			0.131***
			(4.456)			(4.481)			(4.412)
Ifincent	0.030***	0.014***	0.028***						
	(3.010)	(2.625)	(2.829)						
Incrate				0.010**	0.006**	0.010**			
				(2.467)	(2.505)	(2.293)			
Lnvalue							0.002***	0.001***	0.001**
							(2.781)	(3.927)	(2.507)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	3997	3997	3997	3997	3997	3997	3997	3997	3997
Adj_R^2	0.037	0.174	0.042	0.037	0.174	0.041	0.037	0.176	0.041
F	4.442	19.741	4.798	4.373	19.724	4.735	4.411	19.972	4.758
Sobel Z			2.262**			2.187**			2.933***
P value			0.023			0.028			0.003

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.

Source: Created by authors.

Table XIII The impact of investment efficiency

	(1)	(2)	(3)
	F.Medab	*F.Medab*	*F.Medab*
Ifincent	0.035***		
	(3.160)		
Ifincent_Abseffic	-0.220**		
	(-2.175)		
Incrate		0.013**	
		(2.488)	
Incrate_Abseffic		-0.084*	
		(-1.698)	
Lnvalue			0.002***
			(2.614)
Lnvalue_Abseffic			-0.010**
			(-2.018)
Abseffic	0.250***	0.175**	0.232**
	(2.644)	(2.304)	(2.509)
Controls	Yes	Yes	Yes
N	3874	3874	3874
Adj_R^2	0.023	0.022	0.023
F	2.980	2.884	2.918

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.
Source: Created by authors.

4 The tables from Section 5.3

Table XIV Alternative choices for tone management: real earnings management

	Lower level of real earnings management (*Rm*<0)			Higher level of real earnings management (*Rm*>0)		
	(1)	(2)	(3)	(4)	(5)	(6)
	F.Medab	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*
Ifincent	0.035***			0.020		
	(2.678)			(1.331)		
Incrate		0.013**			0.011	
		(2.227)			(1.616)	
Lnvalue			0.002**			0.001
			(2.116)			(1.344)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	2062	2062	2062	1844	1844	1844
Adj_R^2	0.023	0.022	0.021	0.025	0.026	0.025
F	2.061	2.010	1.999	2.139	2.160	2.140

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.

Source: Created by authors.

Table XV Alternative choices for tone management: accrual-based earnings management

	Downward accrual-based earnings management (*Da*<0)			Upward accrual-based earnings management (*Da*>0)		
	(1)	(2)	(3)	(4)	(5)	(6)
	F.Medab	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*
Ifincent	0.031**			0.019		
	(2.104)			(1.391)		
Incrate		0.016**			0.005	
		(2.564)			(0.797)	
Lnvalue			0.002**			0.001
			(2.078)			(0.810)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	1894	1894	1894	1970	1970	1970
Adj_R^2	0.027	0.028	0.027	0.019	0.018	0.018
F	2.234	2.286	2.231	1.850	1.819	1.819

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.

Source: Created by authors.

Table XVI The impact of executive compensation benchmarking

	higher level of executive compensation benchmarking (*Higher Gap*)			lower level of executive compensation benchmarking (*Lower Gap*)		
	(1)	(2)	(3)	(4)	(5)	(6)
	F.Medab	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*
Ifincent	0.036***			0.014		
	(2.809)			(0.949)		
Incrate		0.014***			0.000	
		(2.763)			(0.023)	
Lnvalue			0.002**			0.001
			(2.497)			(0.702)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	2063	2063	2063	1939	1939	1939
Adj_R^2	0.031	0.031	0.031	0.017	0.016	0.016
F	2.516	2.510	2.476	1.728	1.708	1.719

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.

Source: Created by authors.

Table XVII The impact of excessive executive compensation

	Higher level of excessive executive compensation (Higher *Eecomp*)			Lower level of excessive executive compensation (Lower *Eecomp*)		
	(1)	(2)	(3)	(4)	(5)	(6)
	F.Medab	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*
Ifincent	0.030**			0.023		
	(2.280)			(1.550)		
Incrate		0.011**			0.004	
		(2.144)			(0.626)	
Lnvalue			0.001*			0.001
			(1.775)			(1.432)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	2035	2035	2035	1967	1967	1967
Adj_R^2	0.026	0.026	0.025	0.015	0.014	0.015
F	2.275	2.261	2.225	1.702	1.654	1.693

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.

Source: Created by authors.

Table XVIII The impact of negative news coverage

	Lower proportion of negative news coverage (Higher *Newsneg*)			Higher proportion of negative news coverage (Lower *Newsneg*)		
	(1)	(2)	(3)	(4)	(5)	(6)
	F.Medab	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*	*F.Medab*
Ifincent	0.041***			0.008		
	(3.149)			(0.519)		
Incrate		0.016***			0.002	
		(2.782)			(0.334)	
Lnvalue			0.002***			0.000
			(2.825)			(0.227)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
N	2240	2240	2240	1762	1762	1762
Adj_R^2	0.018	0.017	0.017	0.031	0.031	0.031
F	1.921	1.871	1.876	2.281	2.277	2.275

Notes: This table shows the specific path results of transmissions. The value of t-statistics is in brackets; ***, **, * indicate that the parameter estimate is significantly different from zero at the 1%, 5% or 10% level respectively.

Source: Created by authors.

References:

Cai, J. & Walkling, R. A. (2011), "Shareholders' say on pay: Does it create value?", *Journal of Financial and Quantitative Analysis,* Vol. 46 No. 2, pp. 299-339.

Cohen, D. A. & Zarowin, P. (2010), "Accrual-based and real earnings management activities around seasoned equity offerings", *Journal of Accounting & Economics,* Vol. 50 No. 1, pp. 2-19.

Kothari, S. P., Leone, A. J. & Wasley, C. E. (2005), "Performance matched discretionary accrual measures", *Journal of accounting & economics,* Vol. 39 No. 1, pp. 163-197.

Richardson, S. (2006), "Over-investment of free cash flow", *Review of Accounting Studies,* Vol. 11 No. 2-3, pp. 159-189.